Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 1a - Introduction

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Course structure and objectives

Introduction

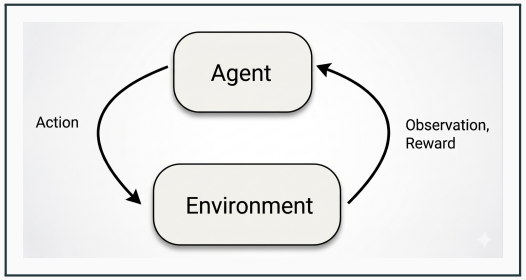
- I am a member of the Computer Science Faculty at the Technion.
- I head the Collaborative AI and Robotics (CLAIR) lab
- My research focuses on multi-agent AI and multi-robot systems.
- My work offers novel approaches to **increasing the capabilities of AI agents via the design of the environments in which they are intended to act**.
- My email: *sarahk@cs.technion.ac.il*

Staff and schedule

- **Teaching:** Sarah Keren
- **TA:** Guy Azran
- **Schedule:**
  - Lesson: Wednesday 14:30-16:30
  - Tutorial: Wednesday 16:30-17:30
  - Reception Hours: per request.

What is this class about?

Sequential Decision Making: deciding how to act now to accomplish a long-term objective



- From observations to decisions
- From low-level to high-level reasoning
- From theory to practice to theory ...
- From single agent to multi-Agent

Our Focus

There is a wide spectrum of AI research questions.

We focus on:

- Non-adversarial settings.
- Single-agent settings.
- Complex and long-term tasks.
- Combining approaches and principled ways to choose the best approach.

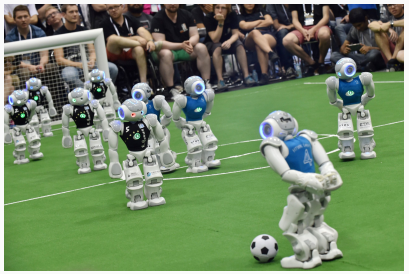
Course Structure

- Weekly 2-hour class followed by a 1-hour tutorial.
- In class, we will cover the theory behind decision-making in AI.
- In the lab, we will practice and test the ideas we discuss in class.
- Grade is set by:
  - 3 homework assignments
  - Quizz (magen)
  - Final exam

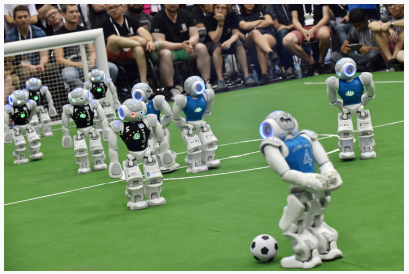
All materials will be available online - participation is non-mandatory.

Decision Making in AI

Reactive vs. Rational Agent



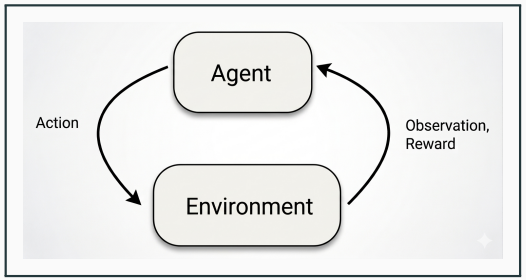
Reactive vs. Rational Agent



- **Reactive** or **Reflex** agent:
  - simple reactions to stimulus (short-term)
  - fast response times and limited resource usage
- **Rational agent:**
  - extended reasoning capabilities
  - can achieve complex objectives
  - utility maximizing
  - can learn new knowledge and react to change

Single Agent Autonomy

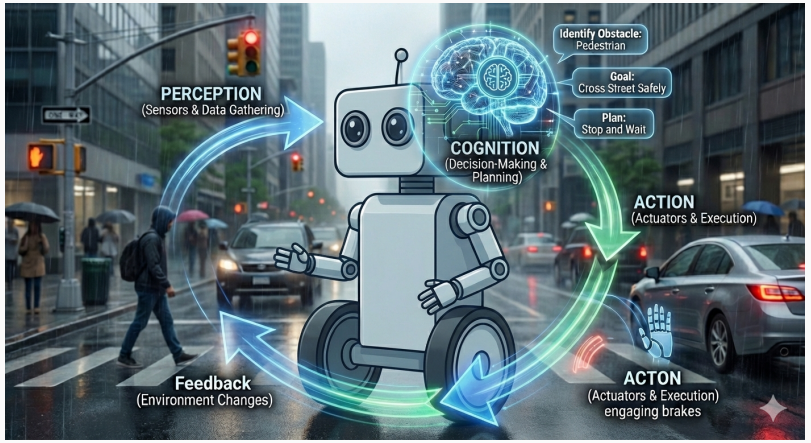
What does it mean to be autonomous?



We seek a **policy** from states (or knowledge about the state) to actions.

Components of an Autonomous Agent?

Action - Perception - Control



Sequential Decision Making

vs.

We will focus on sequential decision making (a.k.a. planning)

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Approaches to Decision Making

We seek a **policy**, which we mark as π that maps states $\mathcal{S}$ or (knowledge about the state) to actions $\mathcal{A}$.

Deterministic $\pi : \mathcal{S} \mapsto \mathcal{A}$ or probabilistic $\pi : \mathcal{S} \times \mathcal{A} \mapsto [0, 1]$

More accurately, we seek a mapping from beliefs $\mathcal{B}$, representing knowledge about the current state (formally defined later on), to actions.

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Basic approaches:

Approaches to Decision Making

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Basic approaches:

- Programming/reactive
- Model-Based Planning
- Machine Learning (from data and experience)
- Reinforcement Learning

Example

You are in Kibutz Ein Dor and you want to get to Austin Texas.

How do you decide how to get to Austin?

Programming/reactive

Very practical in many applications, but lacks flexibility.

Model-Based Planning

Key idea: use the same language for all planning domains.

How can we create the same language for logistics, Mars rover exploration, and cybersecurity?

Model-Based Planning

Characterized by:

- **States** $\mathcal{S}$ a system can be in.
  - A state is a full assignment to the set of **variables** (features) $\mathcal{X}$.
- **Actions** $\mathcal{A}$ that change the current state by changing the values of certain variables.
- **Reward Function** $\mathcal{R}$ that sets a numeric signal passed from the environment (can represent cost)
  - used to signal the objective
  - some domains have **goal conditions** such that the agent should reach **goal states** (e.g., 'be at Austin').
- **Objective:** find a **policy** that drives the initial state to a goal state or that maximizes the expected accumulated reward.

Machine Learning

Use statistics to find patterns in massive amounts of data.

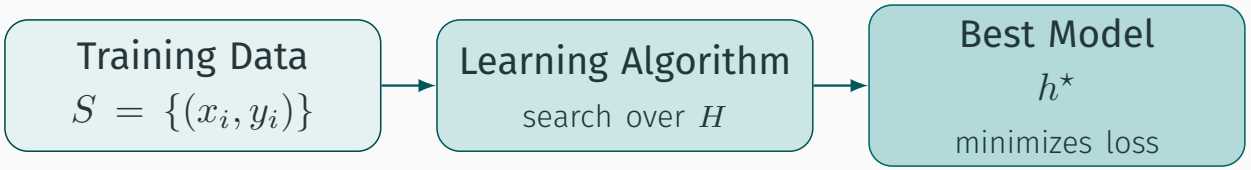
The input vector x (e.g., an image) is mapped to an output $f(x)$, which represents a classification label or a probability distribution over the possible labels. Paradigms of Machine Learning

- **Supervised Learning**
  - Labeled data (x, y)
  - Classification, regression
- **Unsupervised Learning**
  - No labels
  - Clustering, representation learning
- **Reinforcement Learning**
  - Interaction with environment
  - Sequential decisions, delayed rewards

Machine Learning

Template algorithm (for supervised learning):

- **Input:**
  - Sample set: $S = \{(x_i, y_i)_{i=1}^m\}$
  - Model calls H (candidate classifiers)
- **Output:**
  - Classifier with lowest **expected error** (e.g. loss).



Machine Learning: Imitation Learning

- Learning from demonstrations
- Useful when it is easier for an expert to demonstrate the desired behavior rather than to specify a reward function.
- Simplest form: **behaviour cloning (BC)**, learning the expert's policy using supervised learning.

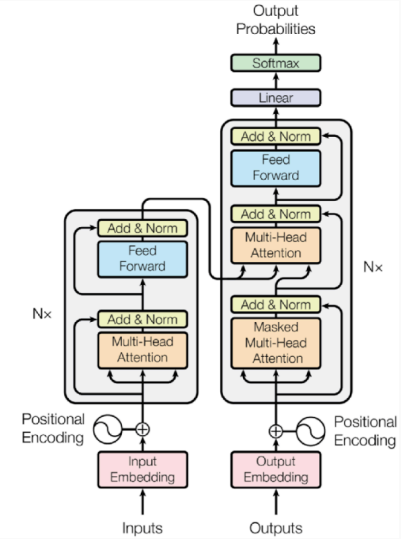
What's the sample set here? What does error (loss) represent?

What Changes with Foundation Models?

In the (not so long ago) past

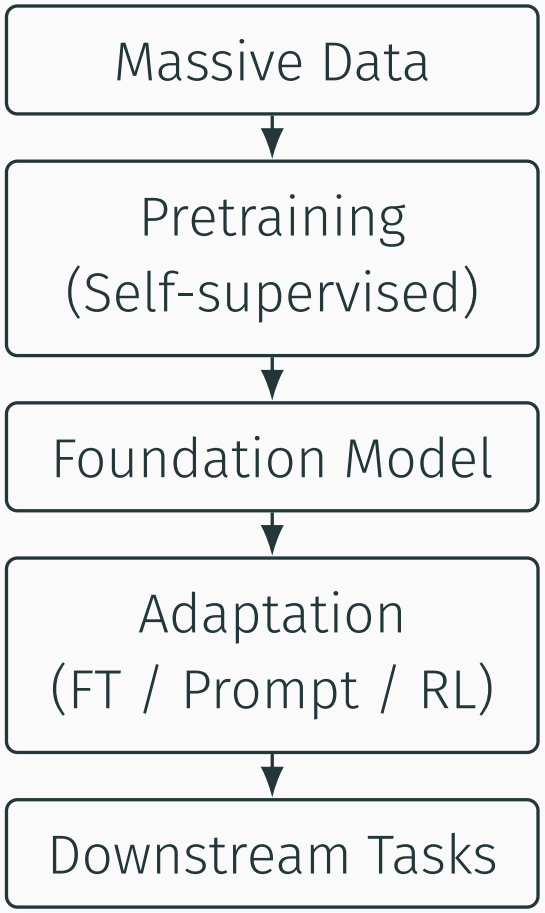
- Train **one model per task**
- Requires labeled data or reward design
- Limited generalization
- Start from scratch each time

With foundation models, we move from task-specific learning to general-purpose models.



Foundation Models: A New Learning Paradigm

- **Core idea:** train once, reuse everywhere
- **Pretraining**
  - Massive, diverse data
  - Self-supervised objectives
  - Learn general representations
- **Adaptation**
  - Fine-tuning, prompting, adapters, RL
  - Task- and domain-specific objectives

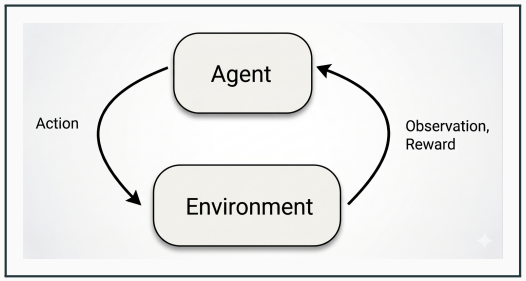


Are We Done?

Even if we have advanced planning, classification, and imitation learning methods - what's missing?

Reinforcement Learning (RL)

- **Learning from interaction** - how to behave to maximize the accumulated reward/achieve a goal.
- The learner is not told which actions to take, but instead **must discover** which actions yield the highest total **reward** by trying them.



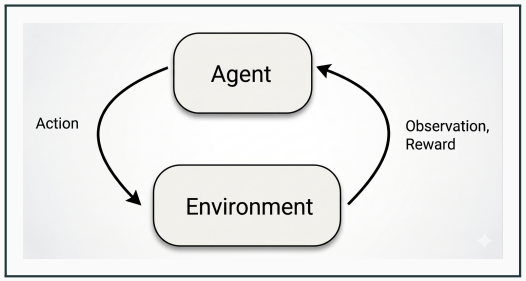
Is reward enough?
https://deepmind.com/research/publications/2021/Reward-is-Enough

Reinforcement Learning (contd.)

The process of Reinforcement Learning (RL) involves:

- Observing the environment
- Deciding how to act using some strategy (policy)
- Acting accordingly
- Receiving a reward (or penalty) and an observation
- Learning from the experiences and refining the policy

The policy is iteratively improved until convergence to an optimal policy.



- RL vs. Planning?
- RL vs. ML ?
- RL vs. IL ?

RL vs. Model-Based Planning

RL vs. Model-Based Planning

Both are fundamental problems in sequential decision-making that aim to find a sequence of actions to achieve a specific goal or maximize utility

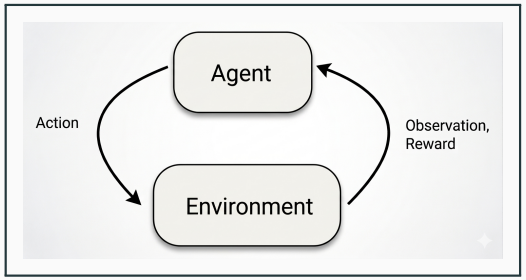
- **Model-Based Planning:** based on a known model of the environment and its dynamics.
- **Reinforcement Learning:** model of the environment is initially not fully known. The agent improves its policy by interacting with the environment.

The distinction is very fuzzy...

RL vs. ML

RL vs. ML

- There is no supervisor, only a reward signal
- Feedback is delayed, not instantaneous
- Date is sequential (non i.i.d): agent's actions affect the subsequent rewards it receives



RL vs. IL

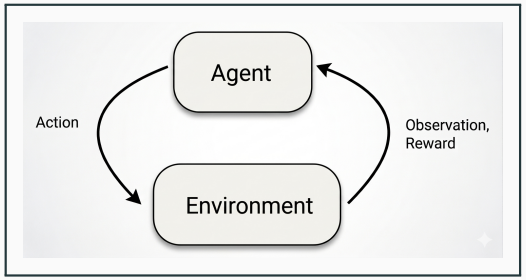
RL vs. IL

Both address sequential decision-making.

- **Imitation Learning (IL):**
  Policy learned directly from expert demonstrations, treating the problem as supervised (or structured prediction) learning, without explicit reward optimization.
- **Reinforcement Learning (RL):**
  Policy learned by interacting with the environment and optimizing a reward signal.

Summary of AI approaches

- Different approaches for decision-making: from data-driven to model-based approaches.
- Different settings require different solutions.



Open Questions:

- How to choose the right paradigm for your problem?
- What is a good model for the decision making process?
- How does the presence of **other agents** affect these decision?